

Grain Size Analysis Report

Report Date: April 20, 2026 at 10:04

Sample Name: Thomson_SERDP_Borden_Sand

Soil Classification: Poorly-graded sand

Temperature: 20.0°C

Porosity: 0.4

Data Points: 15

Executive Summary

Sample Thomson_SERDP_Borden_Sand has been analyzed and classified as Poorly-graded sand.

MEDIAN SIZE (D_{50})

0.180

mm

UNIFORMITY (CU)

2.79

Uniform

CURVATURE (CC)

1.56

Well-graded ($1 \leq C_c \leq 3$)

Grain Size Distribution Analysis

Grain size distribution was determined using sieve analysis and/or sedimentation methods following standard geotechnical procedures. The cumulative distribution curve plots percent passing versus grain size on a semi-logarithmic scale.

Characteristic Diameters

Characteristic grain sizes represent specific percentiles of the grain size distribution:

- **D₁₀, D₃₀, D₅₀, D₆₀:** Grain diameters at which 10%, 30%, 50%, and 60% of the soil mass is finer
- **D₅₀ (Median):** The median grain size, representing the center of the distribution
- These values are fundamental for soil classification and hydraulic conductivity estimation

Gradation Parameters

Uniformity Coefficient: $C_u = D_{60} / D_{10}$

- $C_u < 4$: Uniform gradation (narrow size range)
- $4 \leq C_u < 6$: Moderate gradation
- $C_u \geq 6$: Well-graded soil (wide size range)

Coefficient of Curvature: $C_c = (D_{30})^2 / (D_{10} \times D_{60})$

- $1 \leq C_c \leq 3$: Well-graded with good particle size distribution
- Outside this range: Gap-graded or uniform distribution

Results & Analysis

Characteristic Grain Sizes

D₁₀ 0.070 mm	D₃₀ 0.146 mm	D₅₀ 0.180 mm	D₆₀ 0.195 mm
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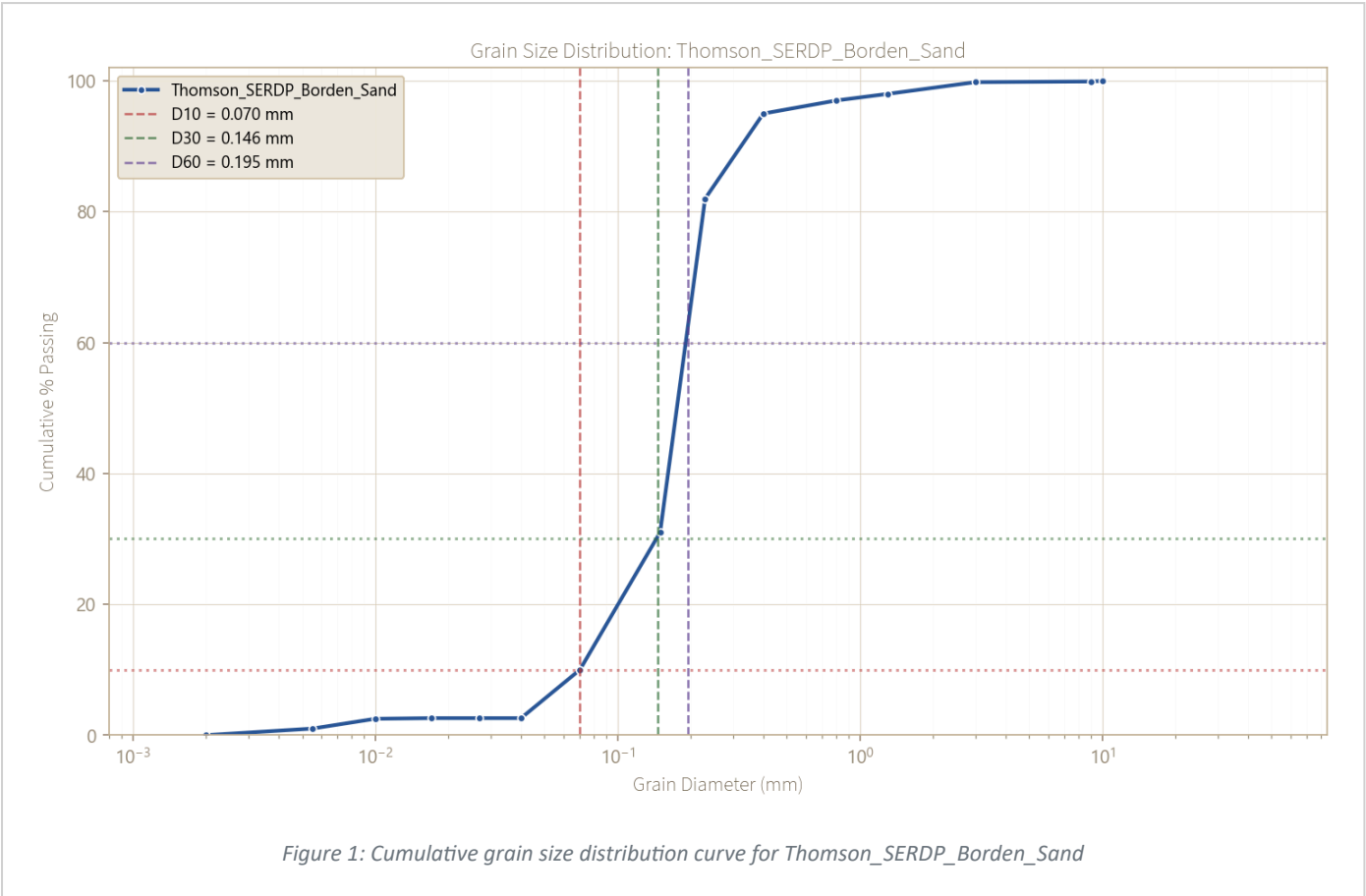
Soil Classification Parameters

Parameter	Value	Classification
Uniformity Coefficient ($C_u = D_{60}/D_{10}$)	2.79	Uniform
Coefficient of Curvature ($C_c = D_{30}^2/D_{10} \cdot D_{60}$)	1.56	Well-graded ($1 \leq C_c \leq 3$)
Soil Classification	Poorly-graded sand	

Gradation Analysis

Fraction	Size Range (ISO 14688 / DS/EN ISO 14688)	Percentage
Gravel	> 2.0 mm	1.1%
Sand	0.063 – 2.0 mm	90.3%
Fines	< 0.063 mm	8.6%

Grain Size Distribution Curve



APPENDICES

APPENDIX A: DETAILED PERCENTILE DATA

Percentile	Size (mm)	Relative (%)
D5	0.050	12%
D10*	0.070	17%
D16	0.093	23%
D20	0.108	27%
D25	0.127	31%
D30*	0.146	36%
D40	0.164	41%
D50*	0.180	44%
D60*	0.195	48%
D75	0.219	54%
D84	0.256	64%
D90	0.335	83%
D95	0.400	100%